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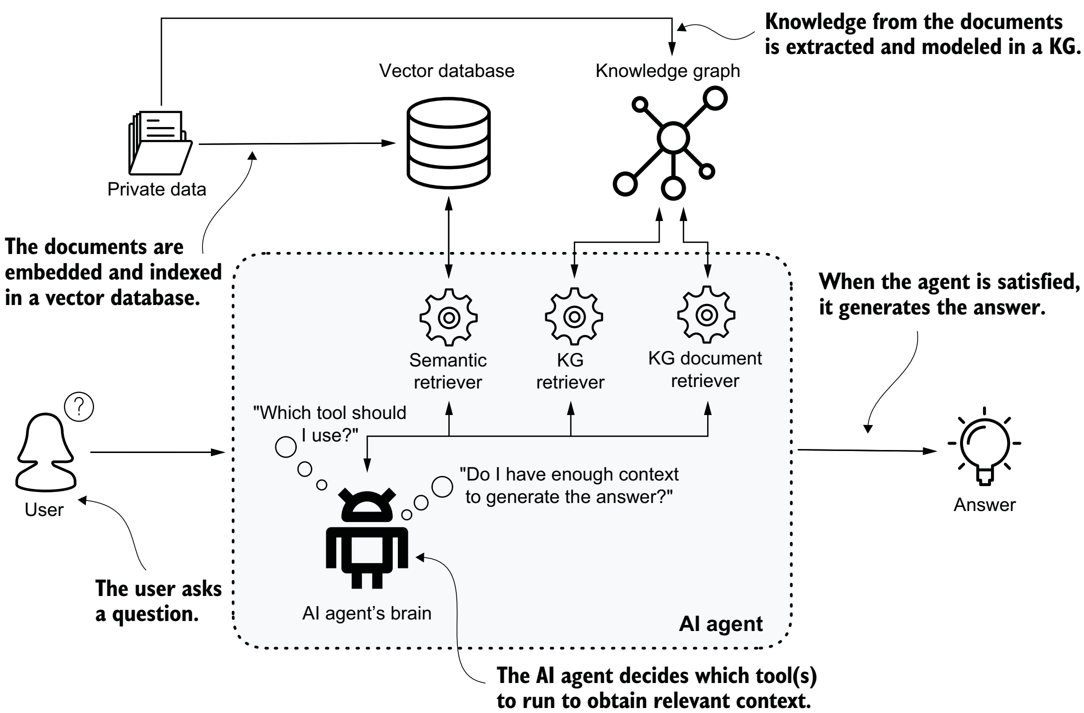
An AI Agent Design for KG-Powered Question Answering



AI/DATA

Knowledge Graphs and LLMs IN ACTION

Negro • Futia • Kůs • Montagna
Forewords by Maxime Labonne, Khalifeh AlJadda

Using knowledge graphs with LLMs reduces hallucinations, enables explainable outputs, and supports better reasoning. By naturally encoding the relationships in your data, knowledge graphs help create AI systems that are more reliable and accurate, even for models that have limited domain knowledge.

Knowledge Graphs and LLMs in Action shows you how to introduce knowledge graphs constructed from structured and unstructured sources into LLM-powered applications and RAG pipelines. Real-world case studies for domain-specific applications—from healthcare to financial crime detection—illustrate how this powerful pairing works in practice. You'll especially appreciate the expert insights on knowledge representation and reasoning strategies.

What's Inside

- Design knowledge graphs for real-world needs
- Build KGs from structured and unstructured data
- Apply machine learning to enrich, complete, and analyze graphs
- Pair knowledge graphs with RAG systems

For ML and AI engineers, data scientists, and data engineers. Examples in Python.

Alessandro Negro is Chief Scientist at GraphAware and author of *Graph-Powered Machine Learning*. **Vlastimil Kůs**, **Giuseppe Futia**, and **Fabio Montagna** are seasoned ML and AI professionals specializing in Knowledge Graphs, Large Language Models, and Graph Neural Networks.

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—Dave Bechberger, Author of *Graph Databases in Action*

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ISBN-13: 978-1-63343-989-4